

The Historical US Funding Advantage Since 1860

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Introduction

- Common belief that US Federal Government has enjoyed a **funding advantage**:
 - Can issue bonds at lower yield than private sector even when bonds have same payouts
 - Measured by **spread** between j -period maturity “plain vanilla”, like-for-like bonds:

$$\underset{\text{Funding advantage}}{\chi_t^{(j)}} := \underset{\text{Yield on private bond}}{\tilde{y}_t^{(j)}} - \underset{\text{Yield on US Govt. bond}}{y_t^{(j)}}, \quad \text{for all maturities } (j)$$

- Many macroeconomists **include** US funding advantage (BIU, BIA, OLG, Bewley):
 - Determines to what extent the government can issue debt unbacked by future surpluses:

$$\text{Market Value (All Govt. Debt)} = PDV[(\text{surplus}_{t+\tau})_{\tau \geq 0}] + PDV[f(\chi_{t+\tau})_{\tau \geq 0}]$$

- **Challenge**: measuring & disciplining χ_t is hard (must match risk, taxes, ...)

This Talk

- **First estimate:** *term structure* of tax/option-adjusted Aaa-Treasury spread (1860+)
 $\chi^{(j)}$ = nominal high grade corporate yield – nominal treasury yield, $\forall$ maturities (j)
- **Revisit equilibrium relationships:** US government funding advantage:
 1. Has been mismeasured and exaggerated in the post-WWI period.
 2. Is negatively correlated with inflation and goes to zero in 1970s-80s (unlike CLP-24)
 3. Very little unconditional correlation with debt-to-GDP (unlike KVJ-12)
 4. Provides little evidence that US strategically exploits monopoly power (unlike CKP-24)

Literature

- Estimation of “convenience spread” on US treasuries
Krishnamurthy and Vissing-Jorgensen (2012), Nagel (2016), van Binsbergen et al. (2022), Koijen and Yogo (2020), Greenwood et al. (2015), Longstaff (2004), Bansal et al (2010)
 - **This paper:** Corrects for maturities, taxes and extends back to mid-19th century
- Estimation of zero-coupon yield curves with Dynamic Nelson Siegel
Nelson and Siegel (1987), Cechetti (1988), Svensson (1995), Dahlquist and Svensson (1996), Andreasen et al. (2019), Diebold and Li (2006), Diebold et al. (2006) and Diebold et al. (2008), Gürkaynak et al. (2007)
 - **This paper:** Bond-specific measurement errors and time-varying pooling
- Historical asset pricing, inflation, and inelastic demand
Daglish & Moore (2018), Choi et al. (2022), Cieslak et al. (2024), Payne et al. (2025), Jiang et al. (2022a), Chen et al. (2022), Jiang et al. (2022b), Jiang et al. (2021b), Jiang et al. (2021a), Koijen and Yogo (2019)
 - **This paper:** Shows that demand elasticity is regulation and policy dependent

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Our Estimates

Stylized Facts About Government Funding Advantage For Macro-Finance

Historical Bond Samples

New Data:

- **New Corporate Bond Data:** prices & features for highest-grade bonds (1860-)
 - Pre-1974: CFC, NYT, Moody's Barron's;
 - Post-1974: Lehman Warga, & Merrill Lynch
- **Treasury Bond Data:** prices & features for all treasuries (1790-)
 - Combines Hall-Payne-Sargent-Szőke data (1790-1940) with CRSP (1926-2024)

Key contribution:

- Identify institutional details that matter for historical bond pricing
- Find relevant bond characteristics and organize bonds accordingly

Substantial Bond Heterogeneity

Tax Advantages for Treasuries:

- **Tax Exemptions (1917-41)**: from federal income taxes on government bonds
- Capital gains are tax advantaged (lower rate, deferred); favors low coupon bonds

Embedded Options:

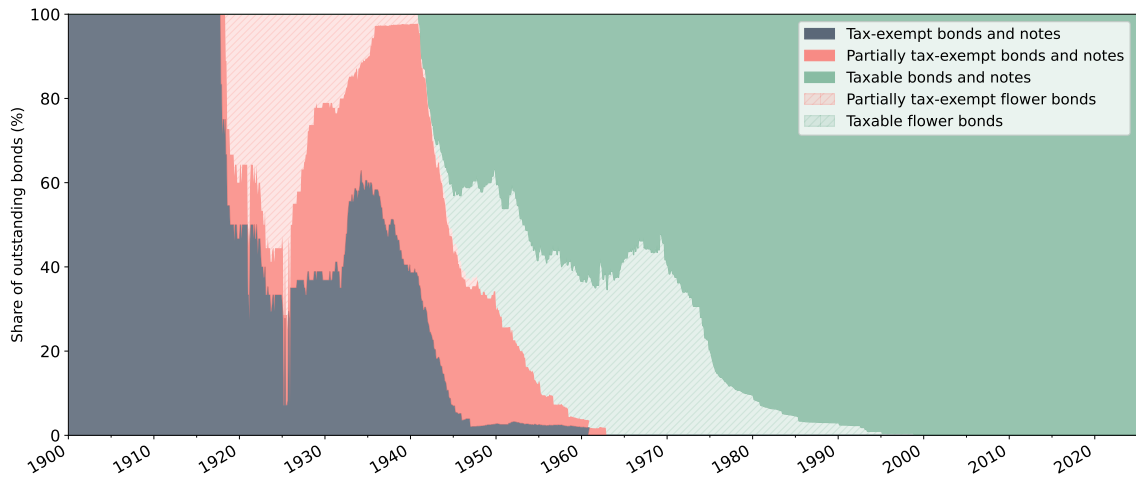
- Call options (on Treasuries & Corporate), Exchange privilege (on Treasuries)
- **Treasury “Flower Bonds”**: Could be redeemed **at par** to pay bondholder’s federal estate taxes upon death:

- Tax provision is valuable when market prices are below par:.
- So effectively acted like an inflation “hedge” or “put”:

$\uparrow \text{Inflation} \Rightarrow \uparrow \text{Interest rates} \Rightarrow \downarrow (\text{Price} - \text{Par}) \Rightarrow \uparrow \text{Value of Estate Tax Discount}$

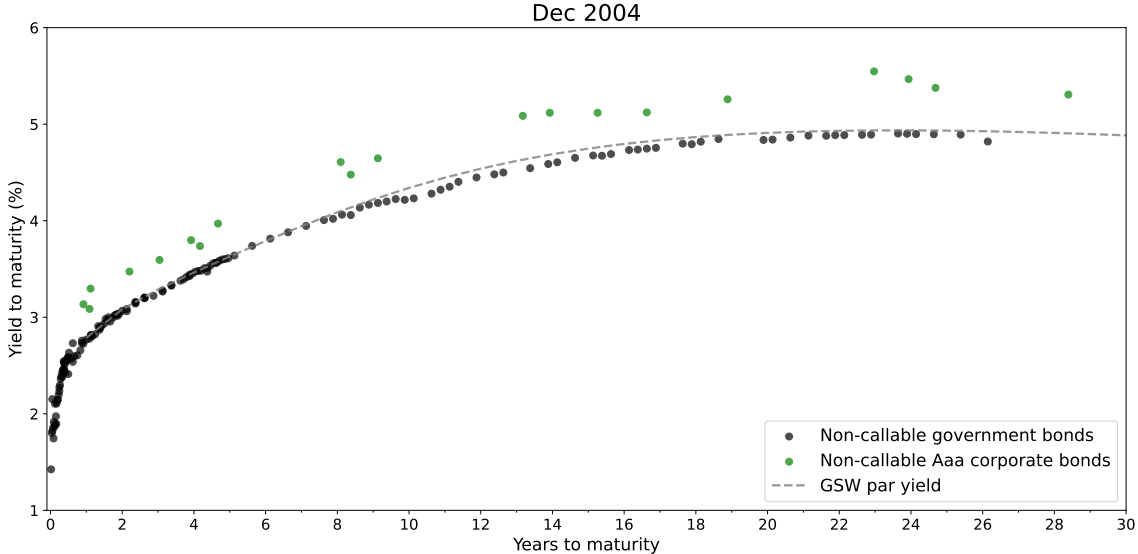
- During 1960-70s, at least half of 10yr+ Treasuries are flower bonds.

Composition of Government Debt: Taxation

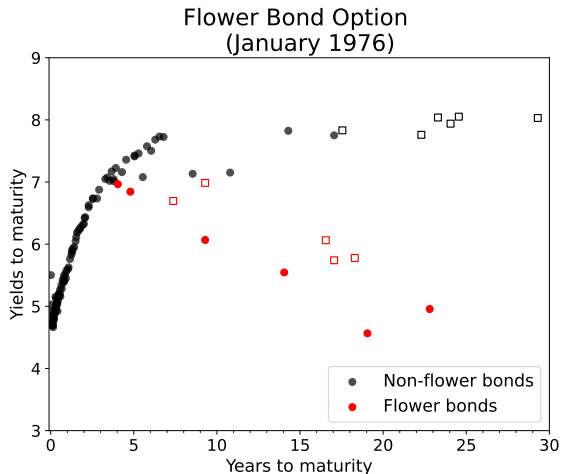
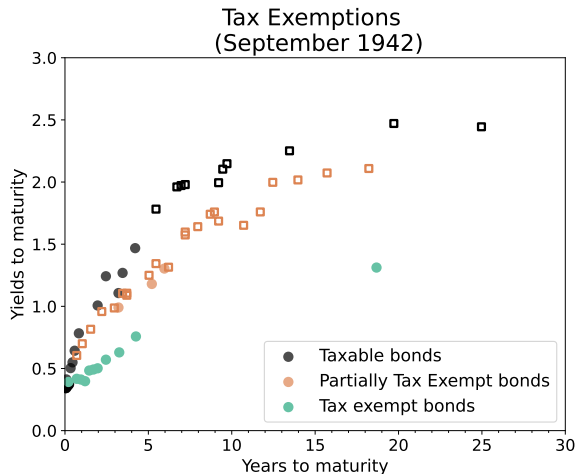
[Callability](#)[Bonds Trading Below Par](#)[Back](#)

Did bond heterogeneity impact prices?

21st Century: Bond Heterogeneity Not a Large Issue



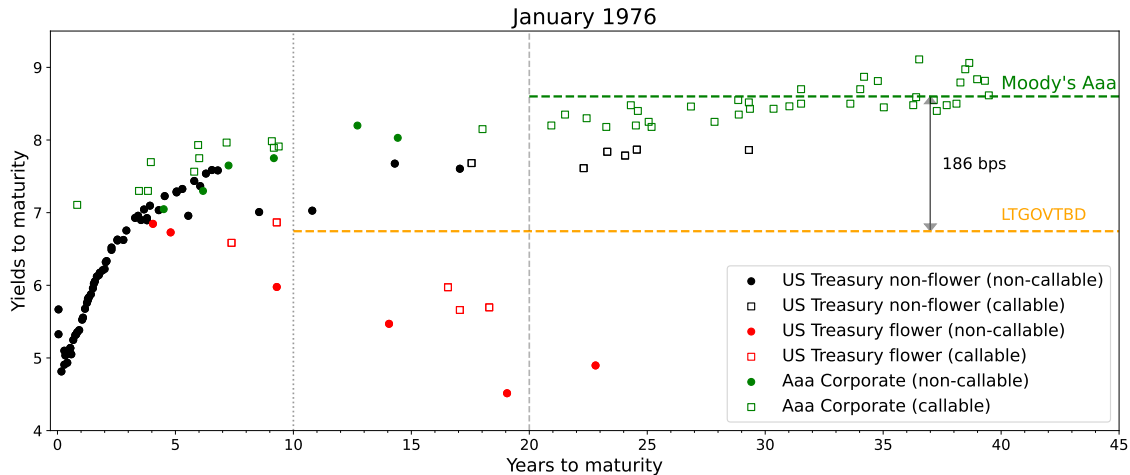
20th Century: Price Distortions From Taxation and Flower Bonds



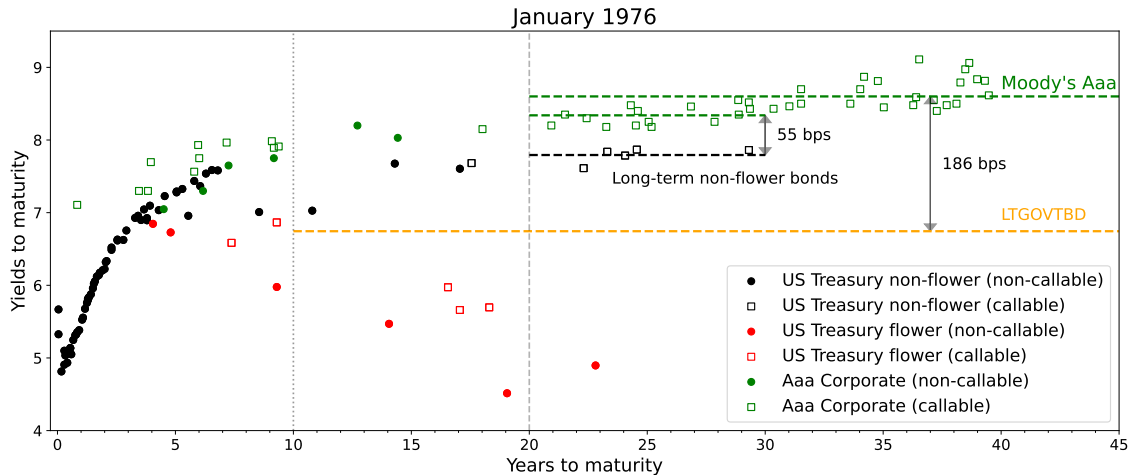
Current Approach: Index-Based Measure of χ_t

- Current papers use the long-term funding advantage series from KVJ-12.
- Computes χ_t as difference between average yield-to-maturity on two bond indices:
 - **Moody's Seasoned Aaa Corporate Bond Yield Index**
(average yield-to-maturity on bonds with maturities ≥ 20 years)
 - The Federal Reserve Board's **Long-Term US Government Bond Yield Index**
*(average yield on **all** outstanding government bonds due or callable after ≥ 10 years)*
- **Implicit assumption:** Bond heterogeneity has negligible price effects (i.e., maturity, tax treatment, options don't matter for bond pricing)
- Let's look at under the hood of these estimates ...

Commonly Used Measure of Long-Maturity Funding Advantage



“Inflation Hedge” in Government Bonds $\Rightarrow$ Mismeasured Spread



How can we make progress on estimation?

Law of one price: common discount function, $\mathbf{q}_t := \{q_t^{(j)}\}_{j \geq 1}$, to price all bonds $i \in I$

$$p_{i,t} = \sum_{j=1}^{M_{t,i}} \underbrace{q_t^{(j)} c_i^{(j)}}_{\text{price error}} + \varepsilon_{i,t}$$

where

- $M_{t,i}$ = maturity,
- $\{c_i^{(j)}\}_{j \leq M_{t,i}}$ = cash flows (coupons and principal)

Identification: simultaneously observe bonds with different maturities and coupons

Yield Curve Estimation

... With Bond Heterogeneity

Law of one price: common discount function, $\mathbf{q}_t := \{q_t^{(j)}\}_{j \geq 1}$, to price all bonds

$$p_{i,t} = \sum_{j=1}^{M_{t,i}} q_t^{(j)} \underbrace{z_i^{(j)}(\eta_t, p_{i,t})}_{\text{tax advantages}} c_i^{(j)} + \underbrace{v_i(\theta_t, p_{i,t})}_{\text{option value}} + \underbrace{\varepsilon_{i,t}}_{\text{price error}}$$

... time-varying wedges $(z_{t,i}, v_{t,i})$ with theory-consistent forms:

$$z_i^{(j)}(\eta_t, p_{i,t}) := f(\text{determinants of tax advantage})$$

$$\approx \underbrace{\exp(\eta_{t,0}^i \hat{y}_{t,i} j)}_{\text{tax exemption}} \underbrace{\exp(-\eta_{t,1} (M_{t,i} - j) \hat{y}_{t,i})}_{\text{tax deferral discount}} \underbrace{\left(1 + \eta_{t,2} \sum_{s=1}^j \max\left\{\frac{\hat{p}_{t+s,i}}{\hat{p}_{t+s-1,i}} - 1, 0\right\}\right)}_{\text{adjustment for capital gains}}$$

implicit capital gains tax advantage

Law of one price: common discount function, $\mathbf{q}_t := \{q_t^{(j)}\}_{j \geq 1}$, to price all bonds

$$p_{i,t} = \sum_{j=1}^{M_{t,i}} q_t^{(j)} \underbrace{z_i^{(j)}(\theta_t, p_{i,t})}_{\text{tax advantages}} c_i^{(j)} + \underbrace{v_i(\theta_t, p_{i,t})}_{\text{option value}} + \underbrace{\varepsilon_{i,t}}_{\text{price error}}$$

... time-varying wedges $(z_{t,i}, v_{t,i})$ with theory-consistent forms:

$$\begin{aligned} v_i^f(\theta_t, p_{i,t}) &:= f(\text{option moneyness, exercise period, interest rate volatility}) \\ &= \exp\left(\theta_{t,0} + \theta_{t,1} \max\{\bar{y}_{i,t} - \bar{y}_{i,t}^p, 0\}\right) M_{i,t}^{\theta_{t,2}}, \quad \text{e.g. for a put option} \end{aligned}$$

Law of one price: common discount function, $\mathbf{q}_t := \{q_t^{(j)}\}_{j \geq 1}$, to price all bonds

$$p_{i,t} = \sum_{j=1}^{M_{t,i}} q_t^{(j)} \underbrace{z_i^{(j)}(\theta_t, p_{i,t})}_{\text{tax advantages}} c_i^{(j)} + \underbrace{v_i(\theta_t, p_{i,t})}_{\text{option value}} + \underbrace{\varepsilon_{i,t}}_{\text{price error}}$$

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Identification: observe bonds with/without options + with/without tax advantages

- Estimate $(\mathbf{q}_t, \eta_t, \theta_t)$ via non-parametric Kernel Ridge (Filipovic-Pelger-Ye-25)

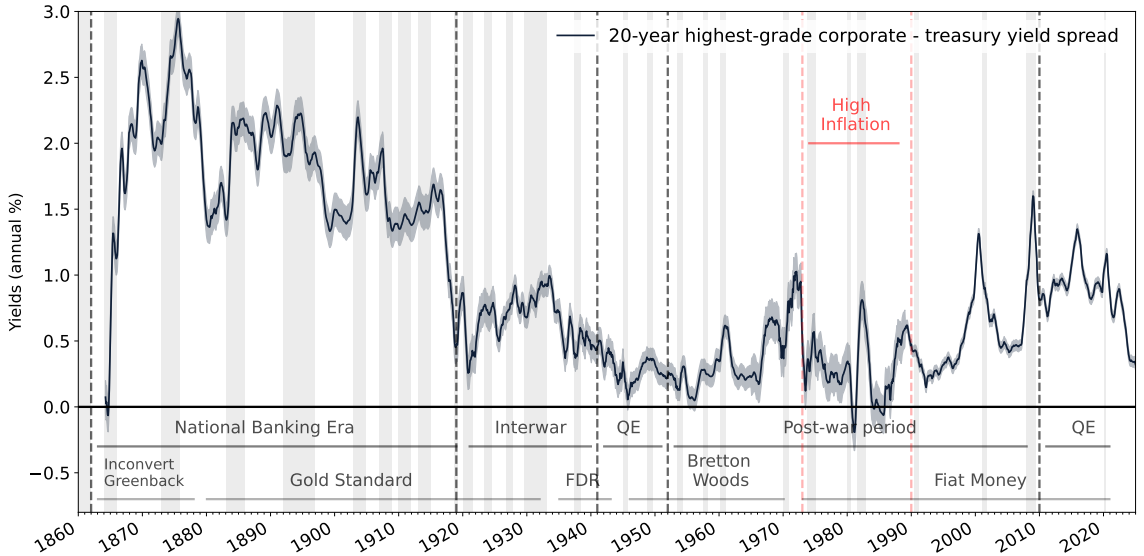
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Methodological Approach For Estimating US Funding Advantage

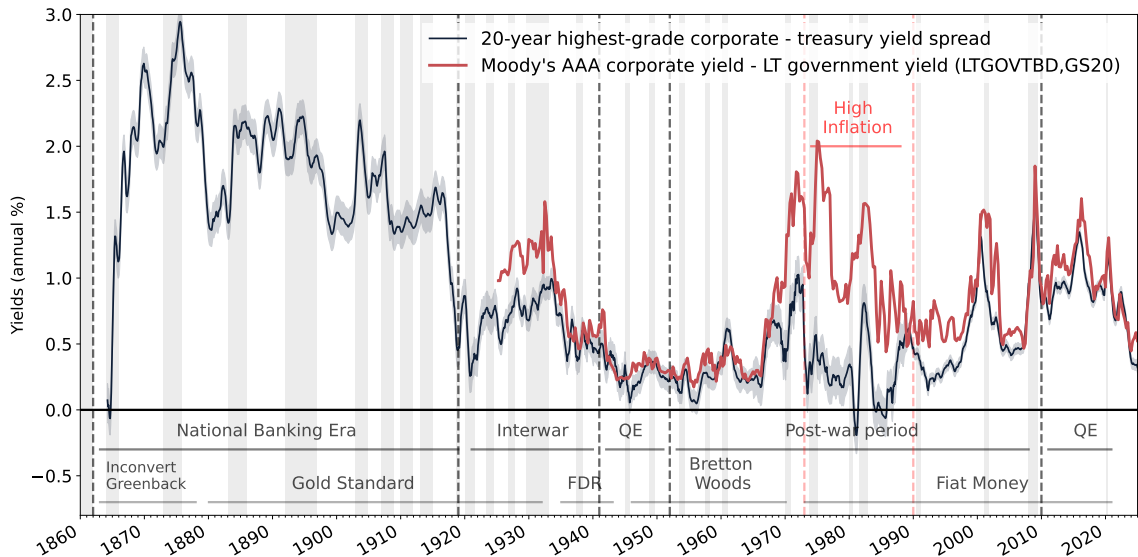
Our Estimates

Stylized Facts About Government Funding Advantage For Macro-Finance

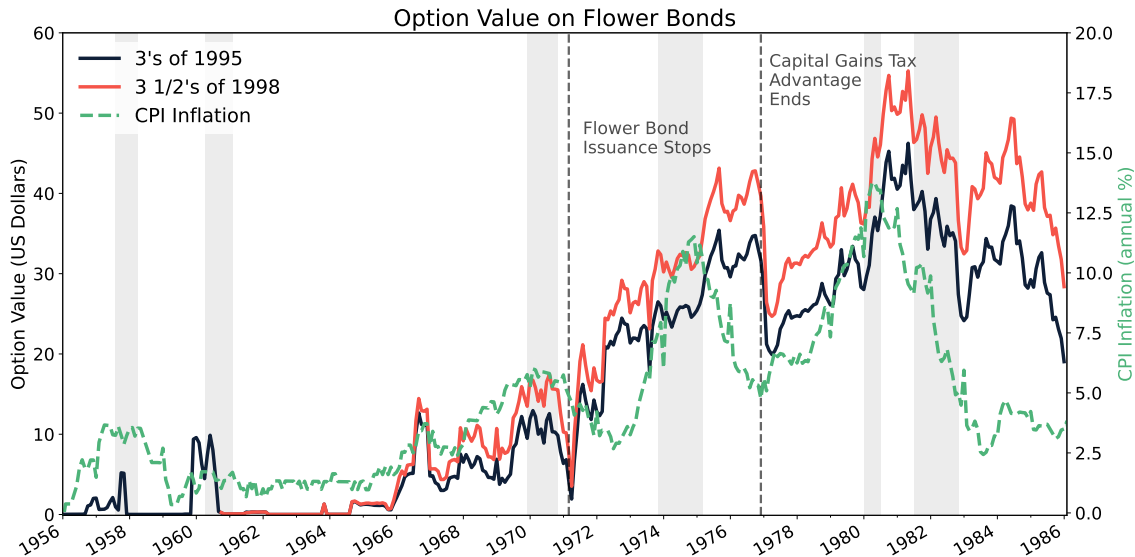
US Funding Advantage 1860-2024



Commonly Used Measure Overestimates US Funding Advantage...

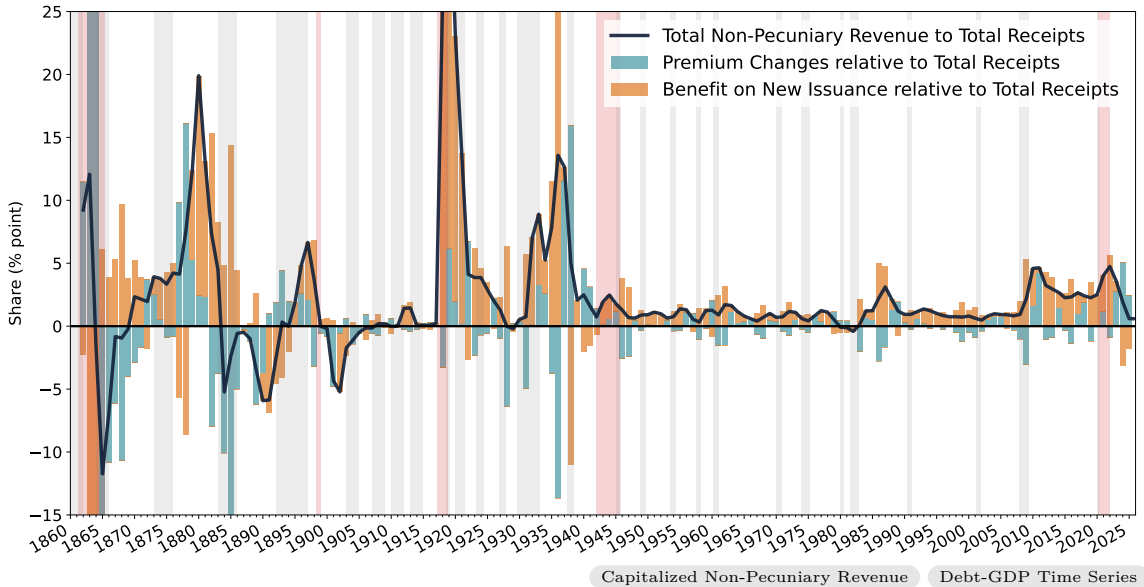


... Because It Includes Options and Tax Advantages



Contribution to the government budget?

Non-Pecuniary Revenue as percentage of Total Federal Revenue



... With Seigniorage Revenue

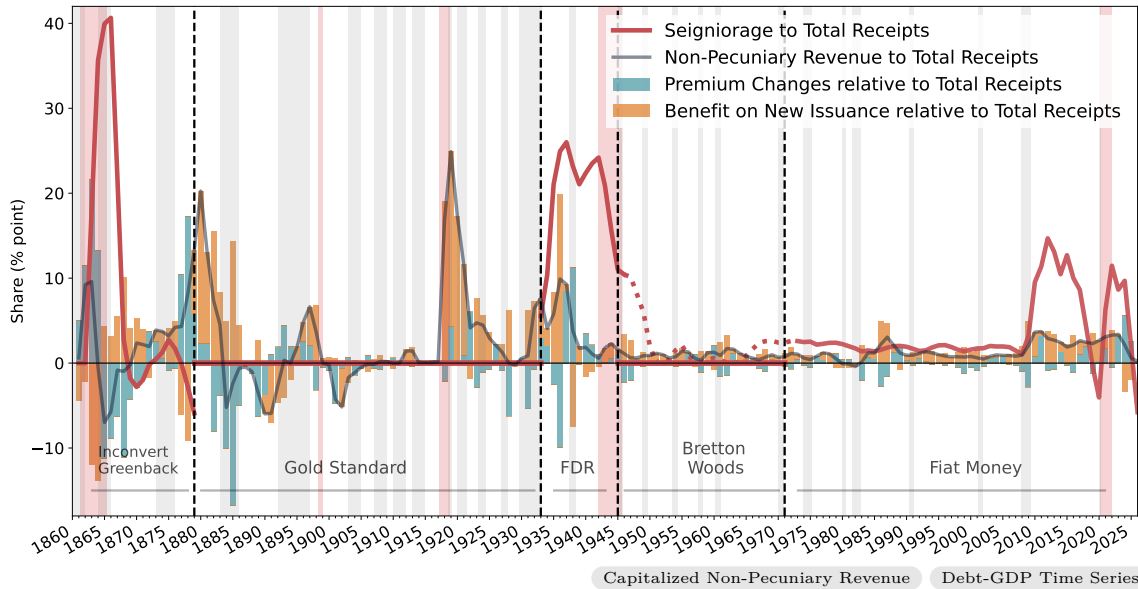


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Stylized Facts About Government Funding Advantage For Macro-Finance

An Asset Pricing Model For The Funding Spread

- Let $\xi_{t,t+1}$ be the pricing kernel for corporate bonds satisfying the dynamic recursion:

$$\tilde{q}_t^{(j)} = \mathbb{E}_t \left[\xi_{t,t+1} \tilde{q}_{t+1}^{(j-1)} \right], \quad j \geq 1, \quad \tilde{q}_t^{(0)} = 1$$

- Let $\Omega_{t,t+1}$ be the non-pecuniary component required to price Treasurys:

$$q_t^{(j)} = \mathbb{E}_t \left[\xi_{t,t+1} \Omega_{t,t+1} q_{t+1}^{(j-1)} \right], \quad j \geq 1, \quad q_t^{(0)} = 1$$

- *Goal:* to understand and (ultimately) model $\Omega_{t,t+1}$.

Long Literature Arguing Quantities Explain Spreads

- Recent work imposes stable equilibrium relationship between spreads & debt-to-GDP (E.g. Krishnamurthy-VissingJorgensen-12, Nagel-16, KL-23, CKP-24, CLP-24)

$$\Omega_{t,t+1} = \exp(\beta_0 + \beta_1 \log(d_t/y_t) + \log(\zeta_t))$$

- d_t is market value of all government debt (+ potentially other “convenient” assets),
 - ζ_t is an independent demand shock.
 - “Micro-founded” using a bond-in-utility model.
- This implies that the funding advantage on 1-period Treasurys is given by:

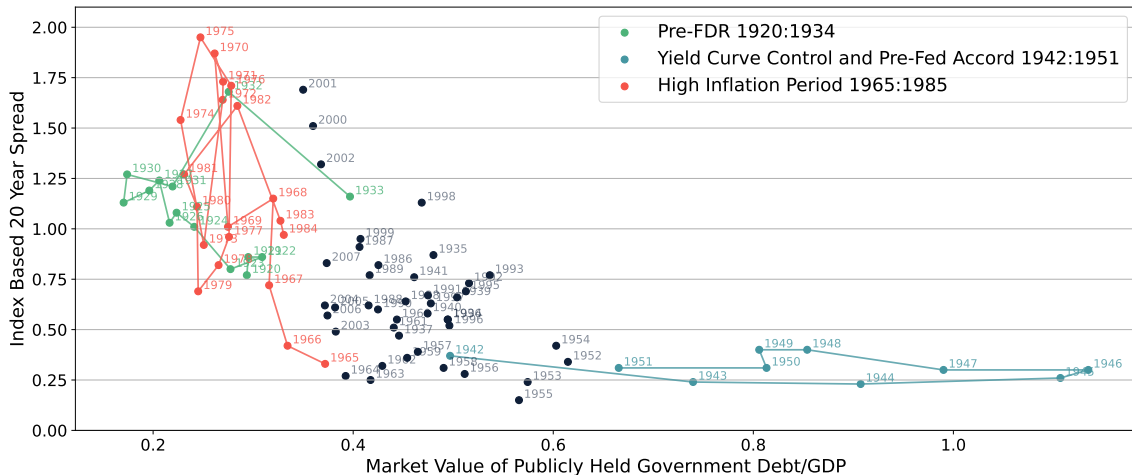
$$\chi_t^{(1)} = \beta_0 + \beta_1 \log(d_t/y_t) + \log(\zeta_t)$$

so “spread” changes are predominately explained by “quantity” changes.

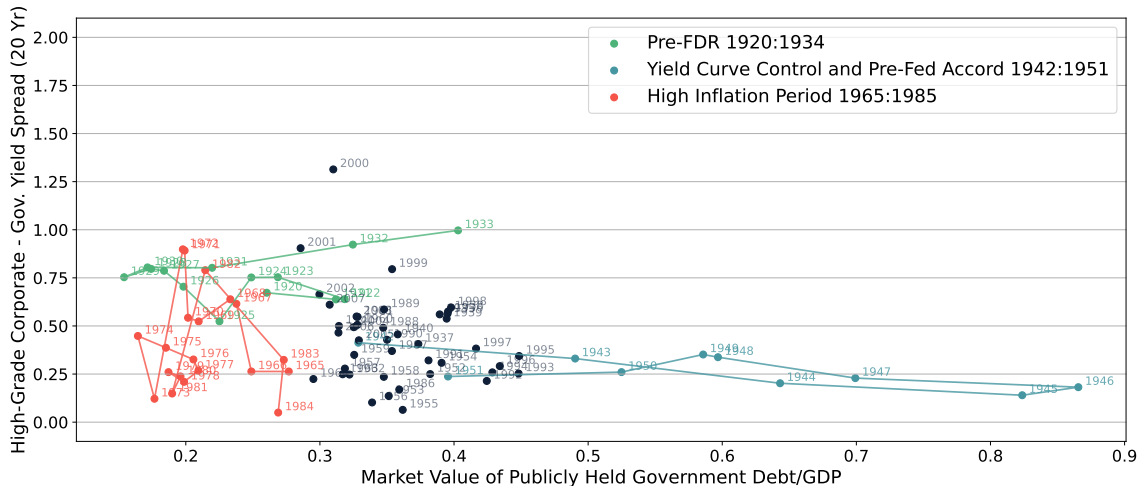
- Relitigates olders debates (e.g. Malkiel (1971), Cook and Hendershott (1978))

Any evidence for this equilibrium relationship?

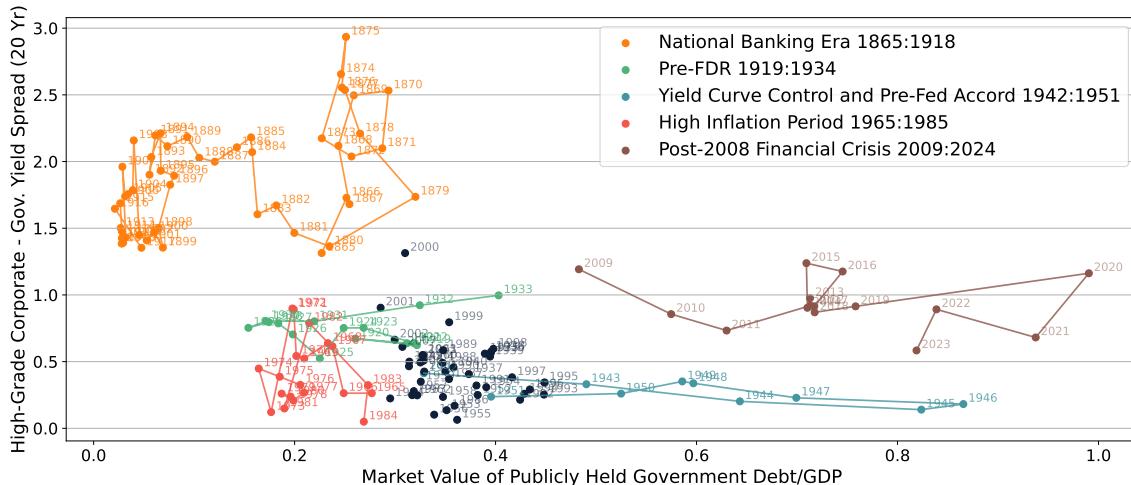
Post WWI: Funding Advantage (20y) vs Debt/GDP: *Index-Spread*



Post WWI: Funding Advantage (20y) vs Debt/GDP: *Our-Spread*



1860-2025: Funding Advantage (20y) vs Debt/GDP



By Maturity

Regressions-KVJ

Regressions-Betas

Instruments

Market Power

Debt-GDP Time Series

Inflation

(After considering tax corrections) . . . perhaps the most striking finding of the study is the lack of support for the view that relative security supplies are a significant factor contributing to the observed spread between the corporate and U.S. yield series studied.

Cook and Hendershoot (1978)

What about an asset pricing model for $\chi_t^{(j)}$?

Asset Pricing Model For The Funding Spread (1920-2007)

- Exponential Affine Model of $\xi_{t,t+1}$ and $\Omega_{t,t+1}$ with a state space $X_t := [\tilde{x}_t, b_t, x_t]$:
 - $\tilde{x}_t$ = Principal components spanning the corporate yield curves,
 - b_t = Principal components of the Treasury's *promised cash-flow matrix* relative to GDP
 - x_t = Residualised principal components of the Treasury yield curves

$$\xi_{t,t+1} = \exp \left(-r_t - 0.5 \lambda_t^T \Sigma \Sigma^T \lambda_t - \lambda_t^T \Sigma \epsilon_{t+1} \right)$$

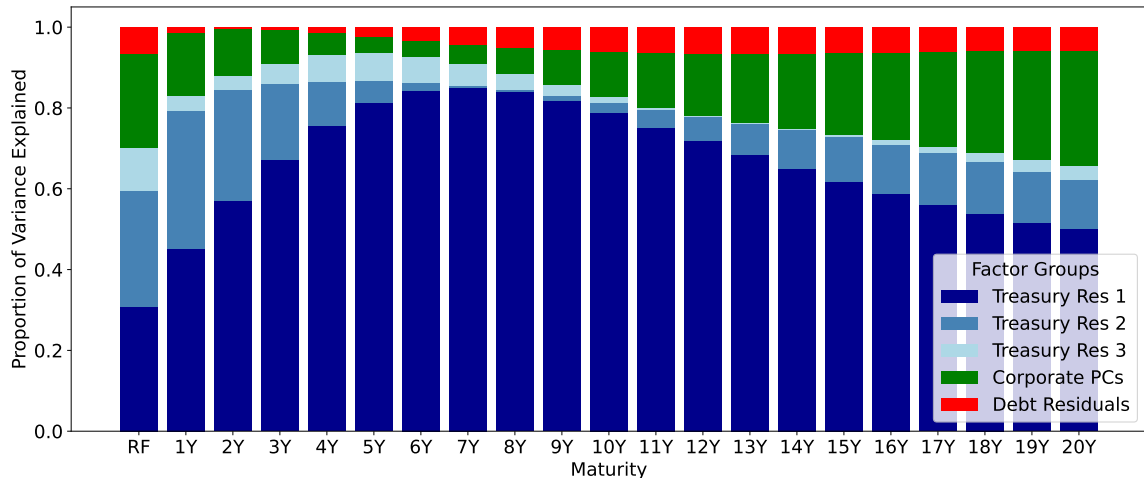
$$r_t = \delta_0 + \delta_1^T X_t \quad \lambda_t = \lambda_0 + \lambda_1^T X_t$$

$$\Omega_{t,t+1} = \exp \left(-\mu_0 - 0.5 \omega_t^T \Sigma \Sigma^T \omega_t - \omega_t^T \Sigma \epsilon_{t+1} \right)$$

$$\omega_t := \omega_0 + \omega_1^T X_t$$

- Estimate using our new term structure of spreads (adapts Adrian-Crump-Moench-13)

Treasury Risk Factors Explain Majority of Variance in The Spread



Key Requirements For Macro-Finance Modeling of Funding Advantage

1. $\chi_t^{(j)}$ does *not* have a stable decreasing, policy invariant relationship to debt supply
...and there are periods (e.g. Civil War, GFC) when $\uparrow$ debt-GDP and $\uparrow \chi_t^{(j)}$ [More](#)
2. Large financial regulation changes coincide with large change in $\chi_t^{(j)}$
...and financial repression (e.g. NBE, GFC) associated with high $\chi_t^{(j)}$
3. Bond risk factors explain most variation in $\chi_t^{(j)}$ within financial regulatory regimes
...and bond-stock betas negatively correlated with $\chi_t^{(j)}$ [More](#)

Regressions-KVJ

Regressions-Betas

Instruments

Debt-GDP Time Series

Inflation

Conclusion

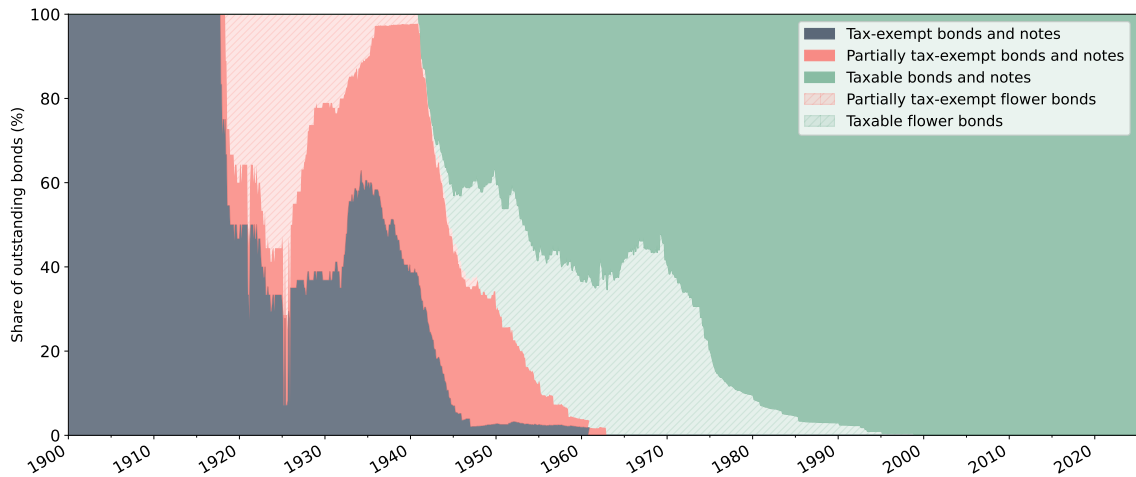
- Tax advantages & embedded options have been misinterpreted as funding advantage.
- It looks like risk pricing matters ... even for the US funding spread.
- So we should focus less on ST debt quantities and more focus on LT debt return risk.
- And think more about tradeoffs when government “choose” $\uparrow$ funding advantage.

Thank You!

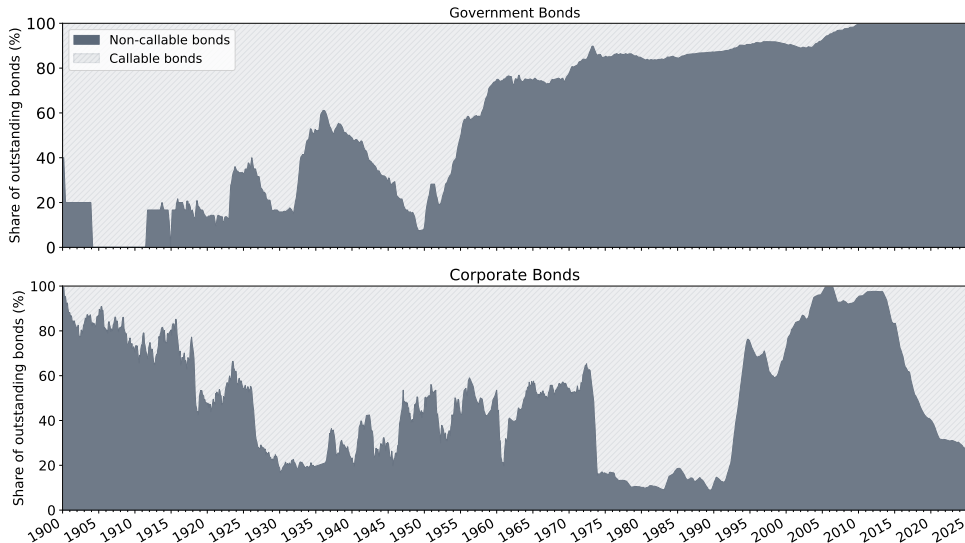
Dataset Spans Many US Financial-Fiscal-Monetary Eras

- *1862-1918*: **National Banking System** of federally regulated one branch banks
 - National banks issued bank notes backed by long term US Federal bonds.
- *1920-1941*: Interwar period (US owns large fraction of world gold supply)
 - Early **Federal Reserve System**: discount window policy (1920s) and QE (1930s).
- *1942-1951*: **Yield curve control**/Quantitative Easing (short/medium yields $\searrow$):
 - Treasury and Fed peg T-Bill rate at 3/8% and cap long rate at 2.5%
- *1952-2007*: Postwar period (“business cycle management”)
 - *1965-1989*: **High-Inflation Period**
- *2010-2023*: **Quantitative Easing** (long yields $\searrow$) and **Dodd-Frank Act**.

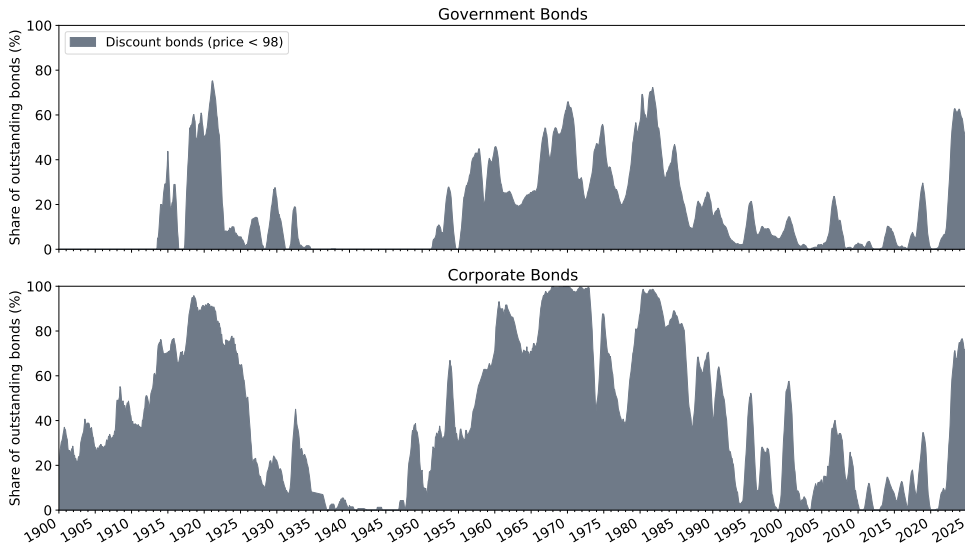
Composition of Government Debt: Taxation

[Callability](#)[Bonds Trading Below Par](#)[Back](#)

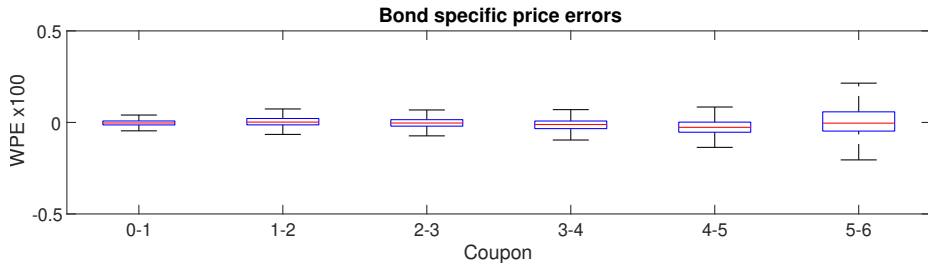
Composition of Government & Corporate Debt: Callability



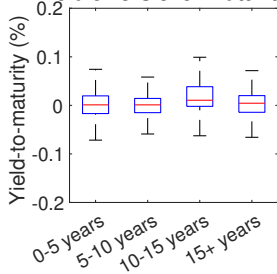
Composition of Government & Corporate Debt: Discount Bonds



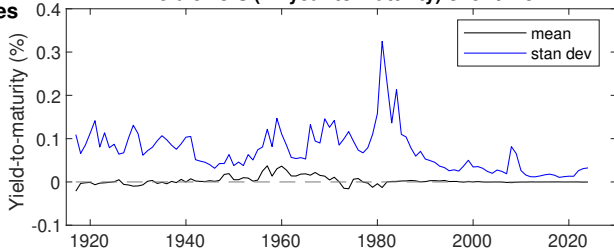
Accounting for Distortions Remove Systematic Patterns in Price Errors



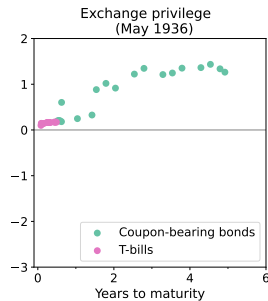
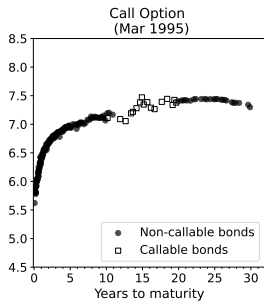
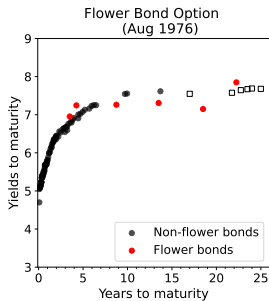
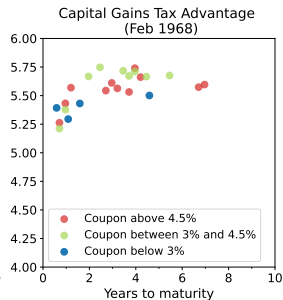
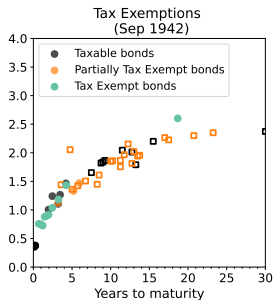
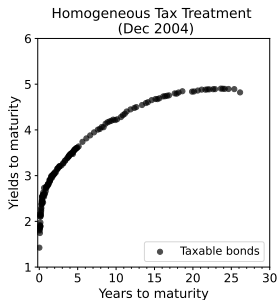
Yield errors over maturities



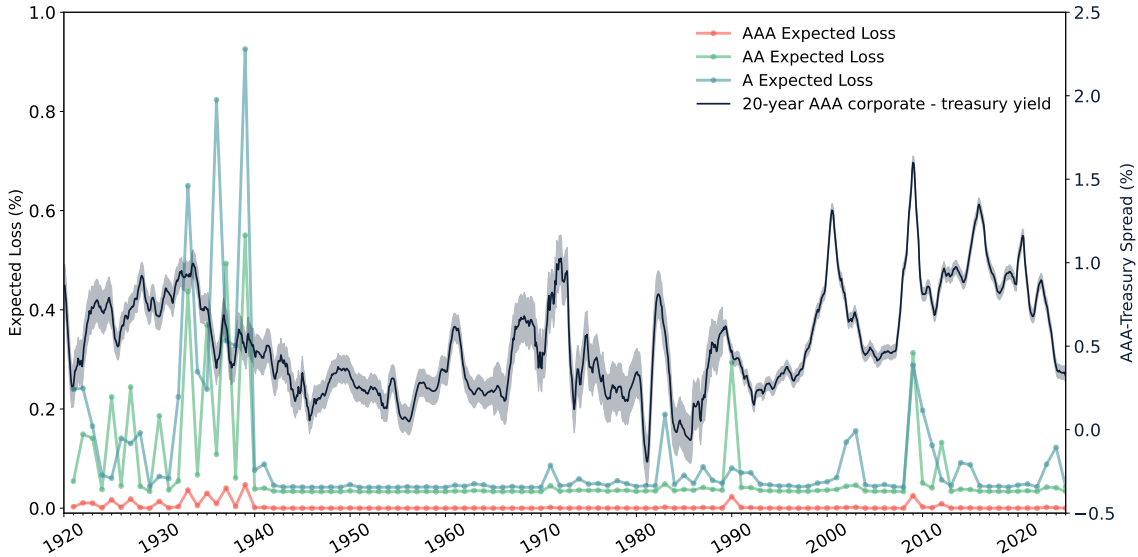
Yield errors (>1 year to maturity) over time



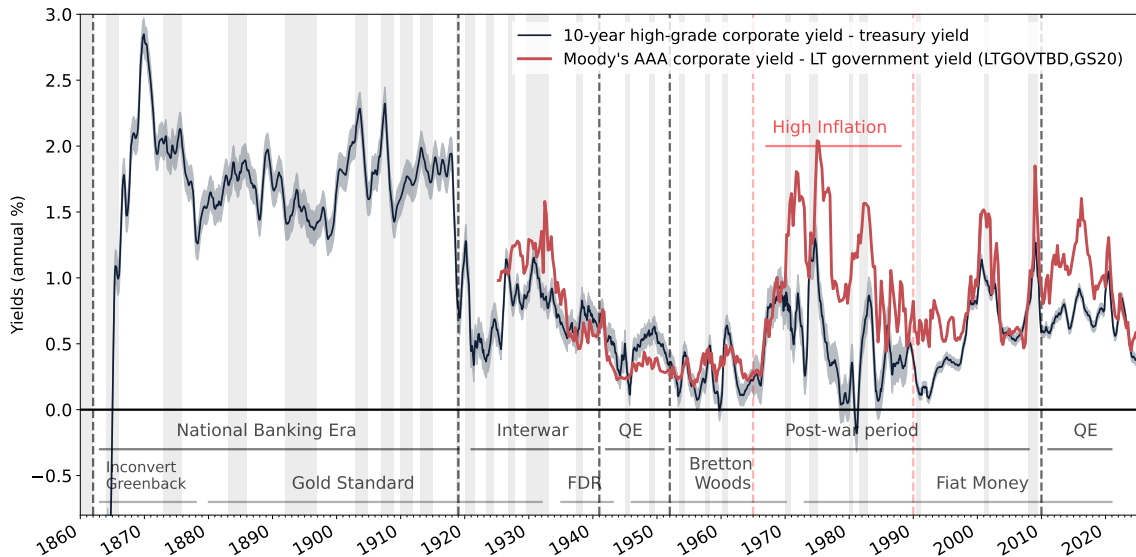
Corrected Yield-to-Maturity Plots



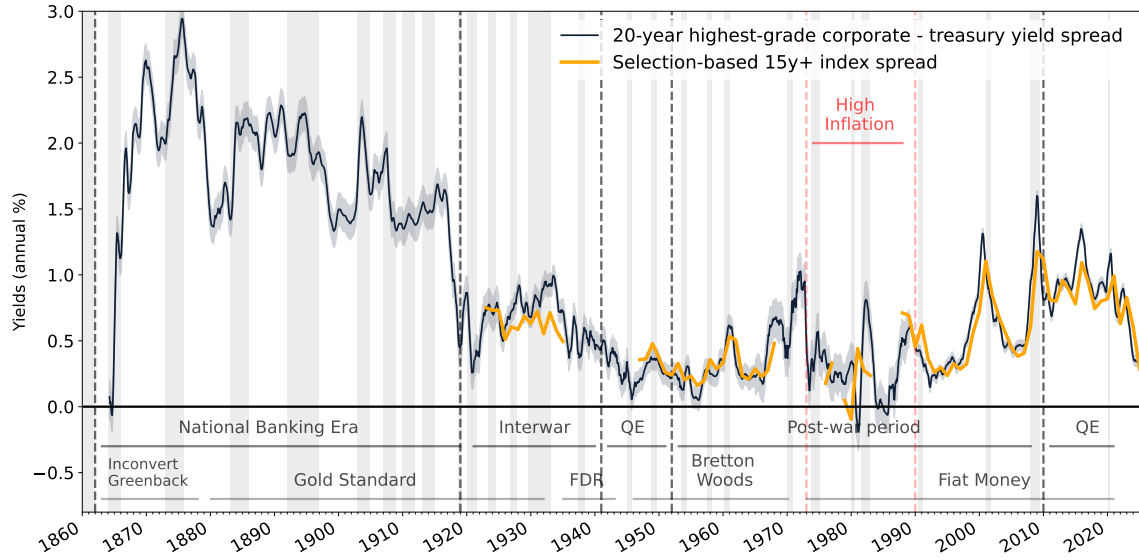
Expected Default Losses



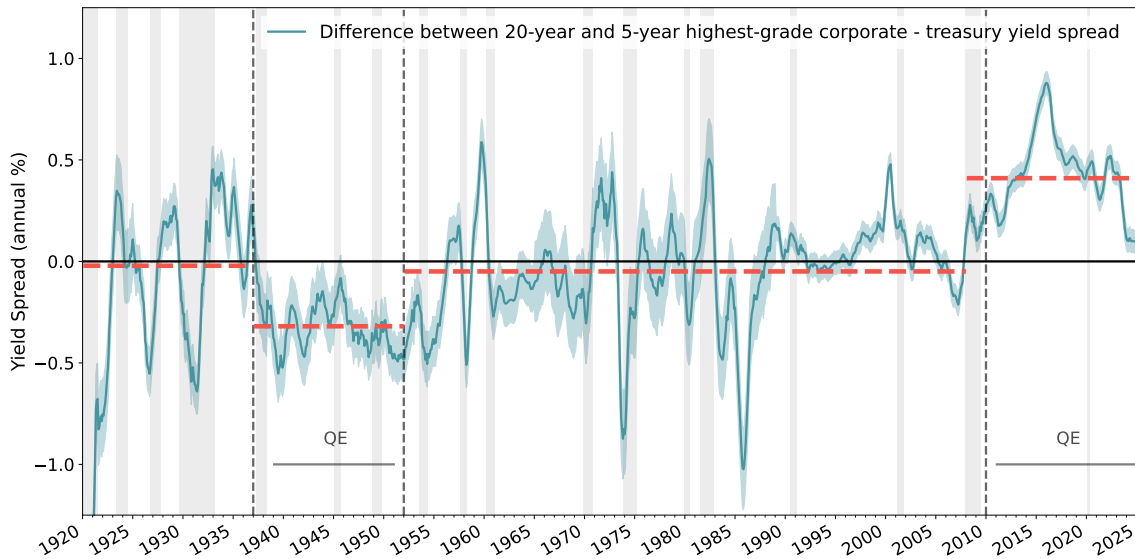
Comparison to Krishnamurthy and Vissing-Jorgensen (2012)



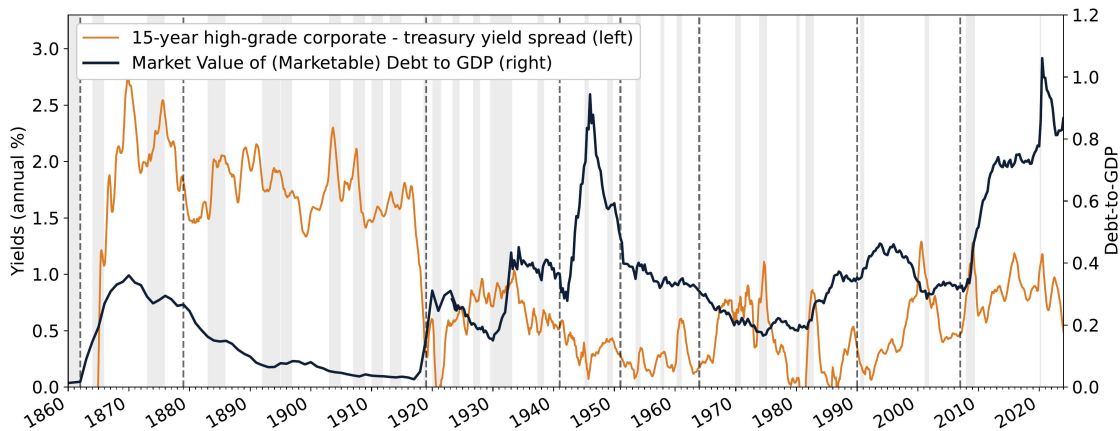
Sense Check: Comparison to “Restricted” Index Spread



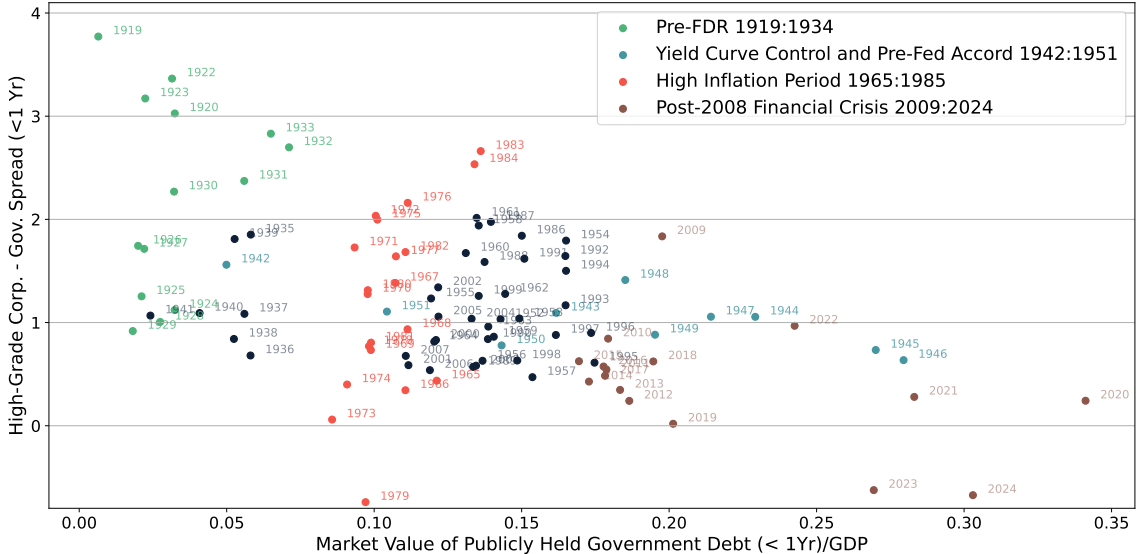
Term Structure Opens Up During QE Episodes



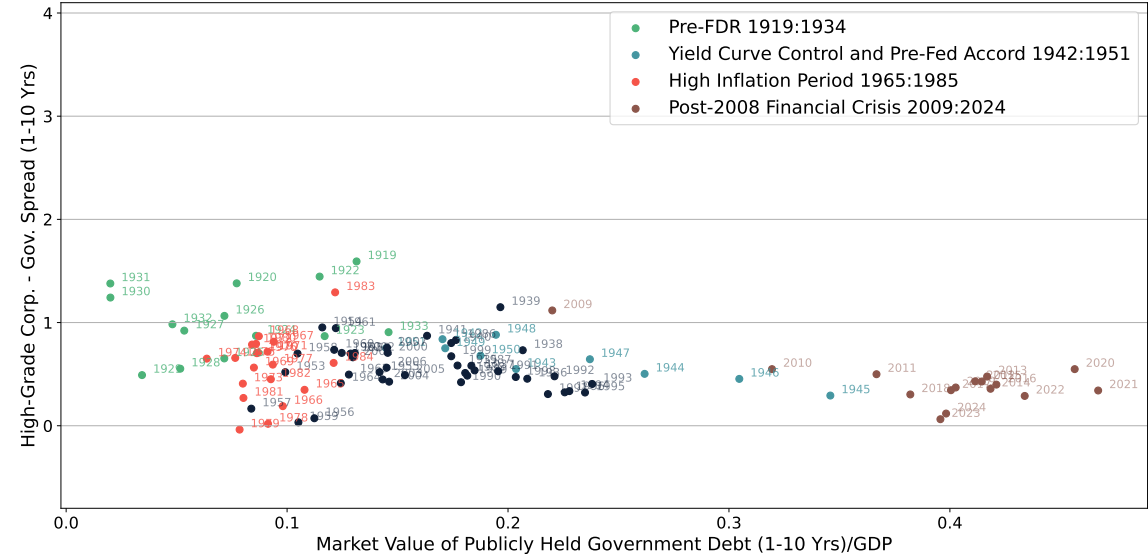
AAA Corporate-Treasury Spread and Debt-to-GDP

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Maturities < 1Yr: Negative Comovement with Debt-to-GDP



Maturities 1-10 Yrs: No Comovement with Debt-to-GDP



Maturities 10+ Yrs: No Comovement with Debt-to-GDP

